Bioresource Technology 400 (2024) 130673

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Case Study

Microaeration promotes volatile siloxanes conversion to methane and simpler monomeric products

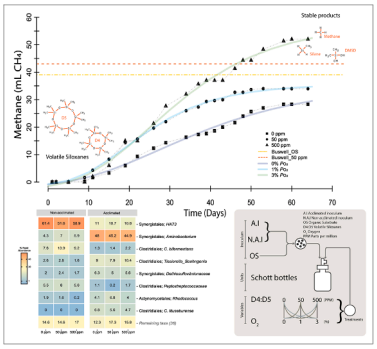
A.E. Ortiz-Ardila [a,b], C. Celis [c], J.G. Usack [d], L.T. Angenent [a,e,f,g], R.A. Labatut [b,*]

[a] Environmental Biotechnology Group, Department of Geosciences, University of Tübingen, Tübingen, Germany
[b] Department of Hydraulic and Environmental Engineering, Pontificia Universidad Católica de Chile, Santiago, Chile
[c] Environmental Technology and Materials Centre, Department of Chemistry, Pontificia Universidad Javeriana, Bogotá, Colombia
[d] Department of Food Science and Technology, University of Georgia, Athens, Georgia
[e] AG Angenent, Max Planck Institute for Biology, Tübingen, Germany
[f] Department of Biological and Chemical Engineering, Aarhus University, Aarhus C, Denmark
[g] The Novo Nordisk Foundation CO_2 Research Center (CORC), Aarhus University, Aarhus C, Denmark

HIGHLIGHTS

- Siloxanes can be converted to methane under anaerobic/microaerobic conditions.
- Microaeration improves the kinetics of siloxane conversion to methane.
- Microaeration modifies the ecological structure of the anaerobic microbiome.
- Previous inoculum acclimatation is not required for siloxane biodegradation.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Anaerobic digestion
Biogas
Siloxane biodegradation
Microbiome adaptation
Emergent contaminants

ABSTRACT

The ubiquitous use of volatile siloxanes in a myriad of product formulations has led to a widespread distribution of these persistent contaminants in both natural ecosystems and wastewater treatment plants. Microbial degradation under microaerobic conditions is a promising approach to mitigate D4 and D5 siloxanes while recovering energy in wastewater treatment plants. This study examined D4/D5 siloxanes biodegradation under both anaerobic and microaerobic conditions (P_{O_2} = 0, 1, 3 %) using wastewater sludge. Results show that the use of microaeration in an otherwise strictly anaerobic environment significantly enhances siloxane conversion to methane. 16S rRNA gene sequencing identified potential degraders, including *Clostridium lituseburense*, *Clostridium bifermentans* and Synergistales species. Furthermore, chemical analysis suggested a stepwise siloxane conversion preceding methanogenesis under microaerobic conditions. This study demonstrates the feasibility of microaerobic siloxane biodegradation, laying groundwork for scalable removal technologies in wastewater treatment plants, ultimately highlighting the importance of using bio-based approaches in tackling persistent pollutants.

* Corresponding author at: Department of Hydraulic and Environmental Engineering, Pontificia Universidad Católica de Chile, Av. Vicuña Mackenna 4860, Macul, Santiago 7820436, Chile.
E-mail address: rolabatut@uc.cl (R.A. Labatut).

https://doi.org/10.1016/j.biortech.2024.130673
Received 9 November 2023; Received in revised form 27 March 2024; Accepted 5 April 2024
Available online 5 April 2024
0960-8524/© 2024 Elsevier Ltd. All rights reserved.

1. Introduction

Volatile siloxanes are used in many industrial and consumer products, particularly in pharmaceuticals and personal care products. In general, siloxane molecules are formed by two organosilane units, which are linked by a silicon-oxygen (Si—O) bond that can be arranged in oligomeric or polymeric forms (Wang et al., 2019). Polymeric siloxanes (e.g., polydimethylsiloxane, PDMS) are arranged as linear molecules with ten or more monomeric Si-O sub-units. Volatile siloxanes are obtained from PDMS hydrolyzation via chemical synthesis, biochemical degradation (Ortiz-Ardila et al., 2021), or chemical rearrangement (Piechota and Buczkowski, 2014). Both forms are considered difficult biodegradation due to the high energy required to break their molecular bonds, particularly the Si—O bond in cyclic molecules, which is also shielded by the steric hindrance of Si-CH_3 groups (Ortiz-Ardila et al., 2023). This bond needs to be cleaved to open the siloxane ring, allowing the degradation of the molecule.

Due to the industrially relevant characteristics of volatile siloxanes, these siloxanes have become almost irreplaceable in the formulation of a variety of industrial products (Ortiz-Ardila et al., 2023). As a result, there has been a considerable increase in their global production, reaching 10 million tons in 2020 (Mojsiewicz-Pieńkowska and Krenczkowska, 2018). However, growth in volatile siloxane production has intensified their negative atmospheric impacts because they are produced faster than they are degraded naturally (Mojsiewicz-Pieńkowska and Krenczkowska, 2018). Among the volatile siloxanes, octamethylcyclotetrasiloxane (D4) and decamethyl-cyclopentasiloxane (D5) are particularly important due to their prevalence in personal care products. D4 and D5 are recognized as emerging contaminants and substances of very high concern by the Evaluation, Authorization, and Restriction of Chemicals regulatory authority in the EU (European Chemical Agency, 2018a,b), following similar regulatory policies in other countries.

D4 and D5 siloxanes are washed down the drain and transported to wastewater treatment plants (WWTPs) – making these facilities the primary concentration points of volatile siloxanes. Due to their high hydrophobicity, most of the volatile siloxanes concentrate in the wastewater secondary sludge. During anaerobic digestion, the D4 and D5 siloxanes contained in the sludge cannot be fully biodegraded, resulting in their release into the biogas. Upon combustion, the volatile siloxanes contained in the biogas cause irreversible damage to the equipment and a significant impact on the capital costs of the WWTP (Struk et al., 2020). WWTPs address these issues using physicochemical methods (e.g., activated carbon adsorption) to clean the biogas (Ortiz-Ardila et al., 2023). However, these methods, although effective at removing the siloxanes, only transfer these compounds to another medium (e.g., biogas to sorbent matrix), requiring further treatment steps before disposal.

To prevent the generation of waste by-products from D4 and D5 biogas removal, their biological degradation has emerged as an alternative (Ortiz-Ardila et al., 2023). However, still there is no clear scientific evidence on how microbes can overcome the steric hindrance in the D4 and D5 molecules (Wang et al., 2019). Furthermore, there is limited knowledge regarding the microbial conversion of D4 and D5 siloxanes into methane. Previous studies have demonstrated that D4 and D5 can be biologically converted to non-volatile final products (i.e., dimethylsilanediol, silanes) under aerobic conditions (Gaj, 2020). However, aerobic treatment systems require permanent air injection and yield up to 12 times more biosolids than anaerobic ones, resulting in high operating costs (van Lier et al., 2023). Nonetheless, anaerobic conditions alone are not sufficient to completely biodegrade D4 and D5 siloxanes due to the lack of a strong electron acceptor (e.g., oxygen) (Zhang et al., 2020).

The controlled addition of oxygen in an otherwise strictly anaerobic environment, known as microaeration, has the thermodynamic benefits of aerobic systems (*i.e.,* stronger electron acceptor) while preserving the ecological relationships in the anaerobic microbiome (Wu et al., 2021). Indeed, previous studies have shown that anaerobic microbiomes can adapt to use oxygen, increasing the metabolic diversity and improving the biodegradation of recalcitrant pollutants (Duarte et al., 2018). However, studies on the use of microaeration for siloxane biodegradation are just emerging. A previous study assessing the biodegradation of D4 suggested that microaeration did not yield considerable changes in the microbiome ecology or a better reactor performance (Santos-Clotas et al., 2020). However, a recent study showed that PDMS were biodegradable and could be transformed into methane under microaerobic conditions once the microbiome acclimated and used oxygen (Ortiz-Ardila et al., 2021).

Previous studies on aerobic D4 siloxane degradation highlight the role of oxygen in overcoming steric hindrance. But do these mechanisms apply under microaerated conditions? Can microbes convert volatile siloxanes into methane? Addressing these questions is crucial for assessing D4 and D5 siloxane microbial conversion feasibility and environmental impact. This study investigates microaeration effects on D4 and D5 biodegradation and methane conversion. Ecological shifts in anaerobic microbiomes under microaeration and siloxane supplementation were characterized, a theoretical reaction pathway for D4 ring opening was proposed, and the hypothesis that both siloxanes and microaeration reshape anaerobic microbiomes to effectively degrade and convert D4 and D5 into methane was tested.

2. Materials and methods

2.1. Experimental design

Biochemical methane potential (BMP) assays were used to assess the biological conversion of D4 and D5 siloxanes to methane under anaerobic and microaerobic conditions. BMPs were performed based on a randomized, full-factorial experimental design testing the following factors: 1) oxygen partial pressure (P_{O_2}); 2) inoculum type; and 3) D4:D5 concentration. The experimental design includes combinations of all the tested factors with triplicates for each combination, with methane production selected as the response variable. Further information about the experimental approach is provided in the supplementary information section (**See supplementary material** Table 1). For factor one, pure oxygen was supplemented at target partial pressures (P_{O_2}) of 1 % and 3 % O_2 for the microaerated treatments, while anaerobic conditions (P_{O_2} = 0 % O_2) were used as a baseline control. For factor two, a fresh, non-acclimated inoculum from an anaerobic digester treating secondary sludge from a municipal wastewater treatment plant (WWTP) (La farfana, Santiago, Chile) was compared to an inoculum from the same digester following acclimation to microaerated conditions and a siloxane-type compound. This inoculum was acclimated to microaerated conditions (P_{O_2} = 1 % O_2) and fed a substrate mixture containing 50 ppm of PDMS for six months *prior* to the BMP assays. For factor three, D4 and D5 siloxanes (Technical grade – Sigma Aldrich) were supplemented in a 1:1 M ratio at a total concentration of 100 ppm (50 ppm of each molecule) and 1000 ppm (500 ppm of each molecule) in the feed mixture. The feed mixture contained glucose, sodium acetate, sodium casein, cellulose, and coconut oil in equal parts, on a chemical oxygen demand (COD) basis as described by Ortiz-Ardila et al. (Ortiz-Ardila et al., 2021). D4 and D5 siloxanes were cooled to 10 °C to reduce their volatility before application to the BMP bottle. Once the D4:D5 mix was dosed, the bottles were gently vortexed to facilitate the transfer of siloxanes to the organic matter (*i.e.,* inocula) inside the bottle.

Finally, four treatments were used as controls and comparisons for each of the experimental combinations: 1) an inoculum without any substrate, to account for endogenous methane production; 2) an inoculum with organic substrate and no-addition of D4:D5, to account for the anaerobic methane production of the organic substrate; 3) same as treatment 2) but under low microaeration, P_{O_2} = 1 % O_2, to account for the effect of oxygen addition in the conversion of organic substrate into methane; and 4) same as treatment 3) but with high microaeration, P_{O_2}

= 3 % O_2. Further information about the experimental design can be found in the supplementary information section (**See supplementary material** Table 1).

2.2. Biochemical methane potential (BMP) assay

The BMP assay was based on Ortiz-Ardila *et al.* (Ortiz-Ardila et al., 2021), using 250 mL Schott bottles at an inoculum-to-substrate ratio of 2:1 (COD basis). Each experimental bottle included the D4:D5 mixture, the organic substrate, and a micronutrient solution (Ortiz-Ardila et al., 2021). Once the bottles were filled, oxygen was removed from the headspace using a flow of nitrogen gas and then placed in a shaking incubator (90 rpm, 37 ± 1 °C). Twelve days into incubation, pure oxygen (99.8 %) was added to each bottle by internal volume substitution with gas-tight syringes to reach the target oxygen partial pressure of 1 % and 3 % $^{v}/_{v}$. Subsequently, oxygen was added to each bottle on biogas measurement days to maintain the target O_2 partial pressures. Finally, the volumetric biogas production was measured from each bottle using the volume displacement method (Ortiz-Ardila et al., 2021). The methane that was produced in the acclimated and non-acclimated inoculum bottles without any substrate was deducted from the methane production on each corresponding experimental bottle. The BMP assay was terminated when the cumulative biomethane production curve reached a plateau phase (*ca.* 55 days).

2.3. Analytical methods

The biogas composition (*i.e.,* O_2, CH_4, CO_2, N_2) was measured using an Agilent 7820A gas chromatograph (GC) (Santa Clara, CA, USA) equipped with a thermal conductivity detector (TCD) and a Carbosphere 1010 capillary column (Supelco-Merck, Germany), according to ASTM method D3612 (American Public Health Association et al., 2017). The lipophilic fraction of siloxanes was recovered from the post-digestion anaerobic sludge by solvent-assisted extraction using an *n*-hexane–acetone mixture (1:1 $^{v}/_{v}$). Also, a tetrahydrofuran extract was obtained and further derivatized using bis(trimethylsilyl) trifluoroacetamide (BSTFA) trimethylsilylating agent, according to Zhang *et al.* (Zhang et al., 2020).

The chemical structure of the extracted compounds was identified using a Shimadzu 8050 gas chromatography-mass spectrometer triple quadrupole (GC–MS-*QQQ*) (Kyoto, Japan). Samples with non-polar compounds, which were both derivatized and underivatized, were injected in splitless mode and separated with a capillary column (HP-5MS, 60 m × 0.25 mm x 0.25 μm, Agilent) with helium as a carrier gas at 1 mL min^{-1}, according to Sanchís *et al.* (Sanchís et al., 2013). The same samples were further analyzed for polar compounds using an Agilent HP-Innowax column (60 m x 0.25 mm x 0.25 μm) based on the methodology reported by Zhang *et al.* (Zhang et al., 2020). Selected ion monitoring (SIM) was used to detect siloxane-like compounds, which were later identified by cross-referencing the NIST14.L mass-spectra library at a 95 % similarity cut-off.

2.4. Theoretical calculations, methane production modeling, and statistical methods employed

The theoretical methane production of each substrate was estimated using the Buswell Formula (B_u) (Symons and Buswell, 1933), which is based on their molecular formula. For D4 and D5, the molecular formulae $C_8H_{24}O_4Si_4$ and $C_{10}H_{30}O_5Si_5$ were assumed. A Gompertz-Zwietering (GZ) model was implemented to assess the kinetic parameters of methane production (*i.e.,* lag phase duration, ultimate methane production, and absolute production rate) for the anaerobic and microaerated conditions (Tjørve and Tjørve, 2017). Model fit was assessed using Akaike's information criterion to reduce model complexity and ensure goodness-f-fit and precision, according to Bonakdari & Zeynoddin (Bonakdari and Zeynoddin, 2022). Average volumetric methane production per treatment was fitted to the GZ model (**Eq. (1)** using non-linear least squares analysis in the stats package for R (R Core Team, 2021).

$$W(t) = A \bullet \exp\left[-\exp\left(\frac{2.718 \bullet K_Z}{A} \left(T_{lag} - t \right) + 1 \right) \right] \quad (1)$$

Where: $W(t)$ = expected value as a function of time, t = time (days), A = upper asymptote (*i.e.,* ultimate methane production [CH_4 ml/mg COD]), K_z = absolute methane production rate (*i.e.,* tangent to the curve), and T_{lag} = lag phase time of methane production curve (days).

Finally, parametric statistical assumptions were tested using the Levenne (homoscedasticity), Kolmogorov-Smirnov (normality), and residual regression model (graphical normality) tests. In addition, all the data were analyzed under a general linear model (GLM), using the test for factorial analysis of variance (2-way ANOVA) to elucidate the interactions of the tested factors. Finally, the HSD Tukey and Scheffé tests were performed as *post hoc* analyses. The statistical tests were conducted using SPSS Statistics 22 software (IBM, Armonk, NY, USA).

2.5. 16S rRNA gene sequencing and taxonomic classification

The 16S rRNA gene sequencing was performed to evaluate the change in microbial community composition before and after the BMP assay. DNA extraction was performed using the FastDNA™ Spin Kit for soil. 16S rRNA gene amplification was accomplished by PCR using 515F and 806R bacterial universal primers with *Illumina* overhang sequences for the MiSeq sequencing platform. As microaerated conditions especially modify the bacterial composition of the AD microbiome, these primers were selected to focus more on bacterial diversity instead of archaeal diversity, according to Hathaway *et al.* (Hathaway et al., 2021). PCR products were cleaned using Mag-Bind PCR Cleanup until a clear, unique band was detected by 1 % agarose electrophoresis. The MiSeq Illumina library was prepared using barcoded indexes Nextera XT Index kit V2 (Illumina Inc. California – US), following manufacturer instructions for dual indexing.

Genomic library products were spectrophotometrically quantified using dsDNA Qubit (Thermo Inc., Waltham, MA, USA), standardized at 4 nM concentration, and pooled for the MiSeq *Illumina* platform. Sequencing was carried out using the MiSeq Reagent Kit v2 (300 cycles) at the Max Planck Institute for Biology, Tübingen, Germany. 16S rRNA gene amplicon sequences were processed *via* Quantitative Insights Into Microbial Ecology 2 (QIIME2) (Bolyen et al., 2019). Sequences were demultiplexed using the QIIME2 default pipeline. Quality filtering, sequence joining, chimera removal, and general denoising were performed using the Divisive Amplicon Denoising Algorithm (DADA2).

The obtained representative sequences were filtered by removing singletons and low-frequency sequences. Next, the filtered sequences were taxonomically classified using the machine-learning Scikit-learn I-Bayes custom-trained classifier. After an initial screening and comparison of the taxonomic classification of the 16S rRNA gene between the SILVA_138 and Greengenes 13_8 databases, the Greengenes database was chosen to train the I-Bayes classifier using the sequence of selected primers (*i.e.,* 515F – 806R). This decision was based on its demonstration of fewer unassigned OTUs and classification results similar to those of the SILVA_138 database. The I-Bayes classifier was then trained, with a cut-off match identity set at 96 % acceptance, using the filtered representative sequences to ensure classification accuracy to at least the taxonomic order level. Finally, unassigned operative taxonomic units (OTUs) were discarded (4 % of initial sequences) to perform α and β diversity analyses.

2.6. Microbial ecology assessment

The α diversity and β diversity metrics were used to evaluate the effect of each experimental treatment on microbial ecology. The α

diversity was determined using the QIIME2 diversity alpha-rarefaction pipeline with Shannon, Simpson, and dominance indices. In addition, a comparative heatmap was created to assess microbial composition in all the experimental conditions. This heatmap was created using the previously filtered sequences and taxonomic classification on the Ampvis2 package (Andersen et al., 2018) under an R-studio environment. The β diversity was determined using the Bray-Curtis distance matrix in a distance-based redundancy analysis (dbRDA). Only significant (ANOVA $p < 0.001$) environmental variables were used as constraints (*i.e.,* oxygen partial pressure and supplemented D4 + D5 concentration), correlating the change in the microbial ecology with the environmental conditions as eigenvectors. Also, a K-means cluster analysis was employed using the type of inoculum (*i.e.,* acclimated, non-acclimated) as a clustering parameter for the calculated Bray-Curtis diversity by the Stats package for R (R Core Team, 2021). Then, using a direct gradient analysis (DGA), significant microbes were ordinated with the constraints of the dbRDA analysis. dbRDA, DGA, clustering calculations, and plots were performed with the Vegan community ecology package for R. The raw fastq files were deposited at the National Center for Biotechnology Information (NCBI) database (Accession number: PRJNA1089110).

3. Results

3.1. D4 and D5 siloxanes are biodegradable and can be converted to methane under anaerobic conditions even without biomass acclimation

Volatile siloxanes (*i.e.,* D4 and D5) were biodegraded and partially converted into methane under all the tested conditions. The theoretical methane yield of the organic substrate mixture is 39 mL CH_4. The final methane yields in all 500 ppm D4:D5 treatments consistently exceeded the 39 mL CH_4 under anaerobic and microaerated conditions (Fig. 1), suggesting that D4, D5, or both were biodegraded to some extent, even though D4 and D5 concentrations were not directly measured during the BMP assay. Likewise, a noticeable improvement in methane production was observed in the 50 ppm D4:D5 treatments compared with the 0 ppm D4:D5 controls under anaerobic conditions (Fig. 1 [A0 and B0]).

Also, treatments with 500 ppm D4:D5 produced 36 % more methane than the control (*i.e.,* no D4:D5 addition) (Fig. 1 [A0]). Similarly, using the acclimated inoculum under the same conditions yielded 60 % more methane than the control (Fig. 1 [B0]). However, there were no significant differences in the final methane yield between non-acclimated and acclimated inoculum treatments under anaerobic conditions (p 0.203; C.I.: 95 %) (Fig. 1 [A0 vs. B0]). These results show that biodegradation of D4 or D5 to siloxane-type compounds is possible without *prior* acclimation. Indeed, previous studies have shown that the biodegradation of D4 and D5 yields silanols and silanes under anaerobic conditions (Santos-Clotas et al., 2019; Wang et al., 2014); however, the mechanisms of converting volatile siloxane into methane have not been elucidated.

The anaerobic treatments using the acclimated inoculum showed increased methane production kinetics, but no effect was observed on the final methane yield. The Gompertz-Zwietering model predicted shorter lag times (T_{lag}) in the acclimated treatments vs. the non-acclimated ones (Table 1). However, non-acclimated treatments showed faster absolute methane production rates (K_z) than acclimated inoculum treatments. This outcome suggests that while acclimated inoculum treatments started methane production earlier, once the lag phase was over, the non-acclimated inoculum treatments were able to convert the substrate mixture more efficiently under anaerobic conditions (Fig. 1 [A0 vs. B0]). These differences and the possible adaptative advantage of the non-acclimated inoculum can be related to the differential ecological composition of both inocula.

The 16S rRNA sequencing results show that the inocula had different compositions in key microbes, which other studies have correlated to recalcitrant compound biodegradation. For example, the abundance of Clostridium bifermentans in the non-acclimated inoculum treatments was higher than in the ones using the acclimated inoculum (Fig. 2 [B]). This microbe has been previously reported to degrade recalcitrant compounds such as: 1) azo dyes (i.e., red 3B-A, black 5, yellow 3G-P) (Joe et al., 2008); 2) tetrachloroethylene (Chang et al., 2001); and 3)

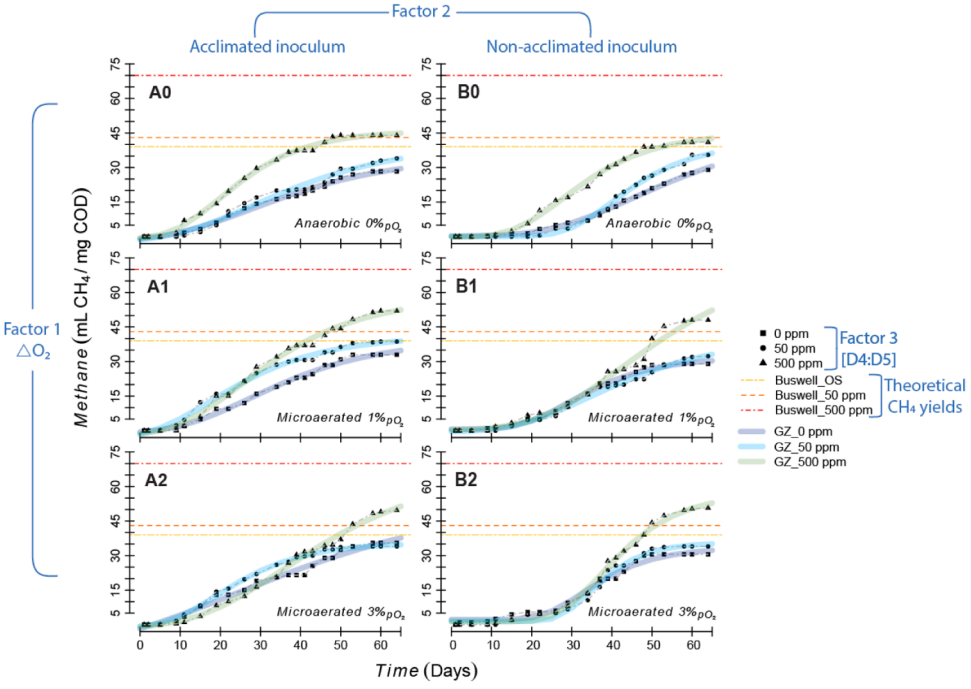


Fig. 1. Measured methane production in BMP assays (mL CH_4/mg COD at STP). Concentrations (pm) refer to the treatments described in the supplementary material Table 1. Plots A0, A1, and A2 correspond to treatments using non-acclimated inoculum. Plots B0, B1, and B2 correspond to treatments using acclimated inoculum. Horizontal dashed lines in the plots correspond to the theoretical methane potential of the organic substrate and D4:D5 siloxanes based on the Buswell formula. Bold lines correspond to the Gompertz- Zwietering (GZ) model fitted to the data. Model equations and parameters can be found in Table 1.

Table 1
Gompertz-Zwietering model (**Eq.** (1)) fitted to the observed methane production of the BMP assays. AIC refers to the Akaike Information Criterion of the model fit to the data.

Inoculum	Fig. 1 Plot	Treatment	Ultimate CH_4 production (A) [CH_4 mL/mg COD]	Absolute CH_4 production rate (K_z)	Lag phase time of CH_4 production curve (T_{lag}) [days]	AIC (%)
Non-Acclimated	A0	C1_N	35	3.1	6.7	71
		T1_N	41	3.6	7.4	94
		T2_N	46	6.8	7.7	74
	A1	C2_N	40	3.4	9.2	73
		T3_N	41	4.9	1.9	81
		T4_N	58	6.2	8.3	92
	A2	C3_N	55	3.3	11.6	93
		T5_N	36	5.2	6.6	55
		T6_N	67	5.4	13.7	86
Acclimated	B0	C1_A	45	3.7	24.2	52
		T1_A	41	6.1	28.6	65
		T2_A	45	6.2	15.1	77
	B1	C2_A	32	4.5	17.1	67
		T3_A	40	3.8	15.7	76
		T4_A	74	5.9	21.2	106
	B2	C3_A	34	5.4	22.6	92
		T5_A	36	7.8	26.9	94
		T6_A	57	8.8	26.5	94

cyclic nitroamines (Zhao et al., 2004) under anaerobic conditions. Furthermore, the distance-based redundancy analysis (dbRDA) showed that C. bifermentans was significantly correlated with the biodegradation of D4 and D5 siloxanes under anaerobic conditions in the non-acclimated treatments (Fig. 3).

The microbiome composition also revealed a significantly higher abundance of general heterotrophic populations (*i.e.,* generalists), namely Peptostreptococcaceae spp. and Clostridiales in the non-

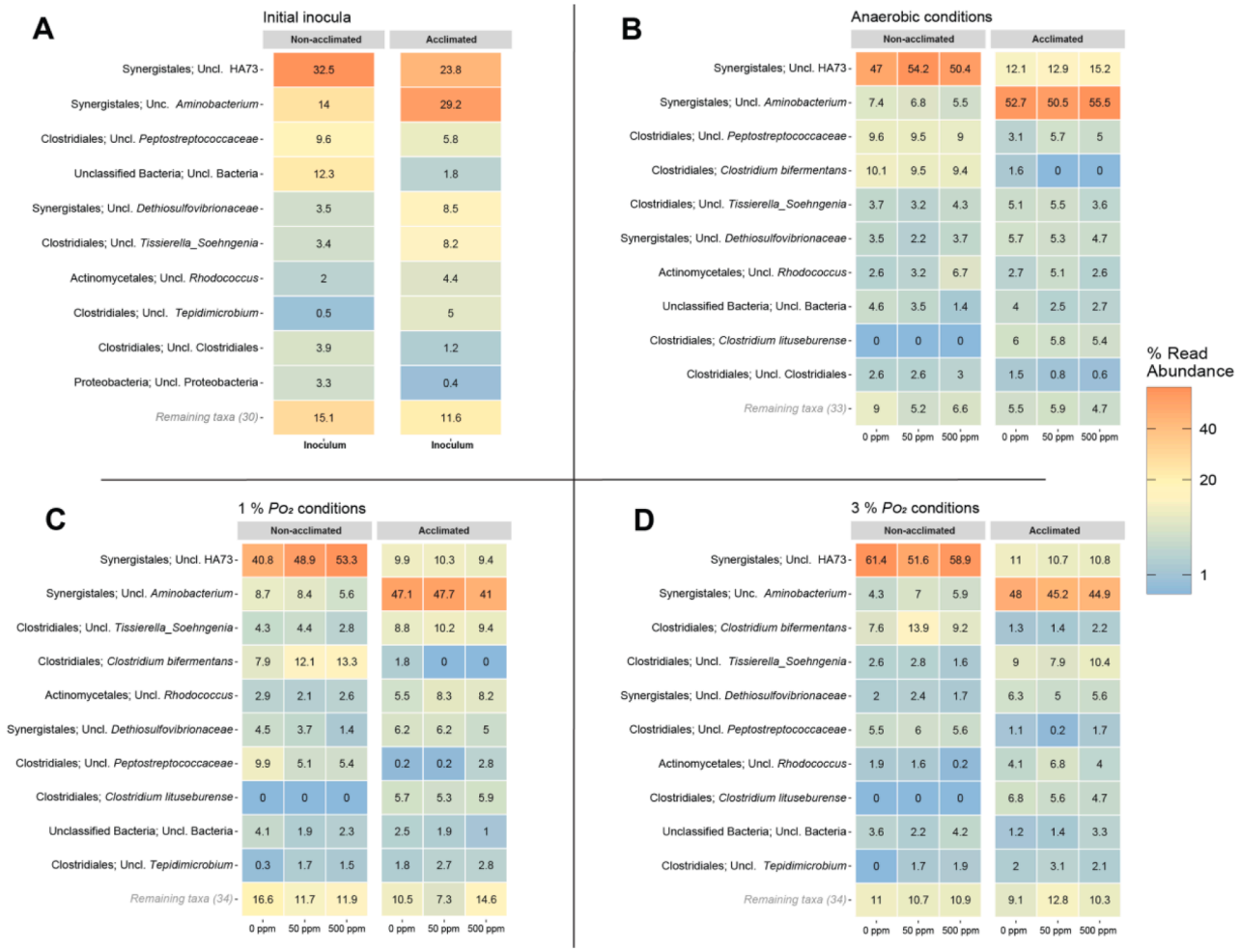


Fig. 2. Microbial distribution heatmap for the ten most abundant taxa per type of inocula at different conditions of the BMP assays. A) Acclimated and non-acclimated inocula before the BMP assay. B) Anaerobic conditions. C) Microaerobic conditions, 1% P_{O_2}. D) Microaerobic conditions, 3% P_{O_2}. Numbers in the heatmap correspond to the % relative abundance of each taxon. Values in the lower axis correspond to the D4:D5 concentrations used in the treatments.

acclimated inoculum treatments. These microbes have been associated with the hydrolytic stages of secondary sludge anaerobic digestion (Militon et al., 2015), the source of the inoculum used in this study. The increased abundance of generalists (*i.e.,* Peptostreptococcaceae spp. and Clostridiales) and specialist (*i.e.,* C. bifermentans) populations in the non-acclimated inoculum treatments may explain the differences in lag phase time and methane production rates (K_z) with the acclimated inoculum treatments. Treatments using non-acclimated inoculum had slower lag phases as specialist populations typically grow slower than generalist ones (Xu et al., 2022). Once C. bifermentans reach higher abundances, they might interact with the generalist's populations, increasing the hydrolysis of the supplemented substrate and explaining the higher methane production rate (Kz) on the non-acclimated inoculum treatments compared with the acclimated one.

On the other hand, the microbial composition of treatments using acclimated inoculum showed a lower abundance of Peptostreptococcaceae spp. and Clostridiales hydrolytic populations (Fig. 2 [B]). However, these microbiomes showed a slightly higher abundance of Tissierrella_Soehngenia spp., another generalist group, which has been associated with the hydrolytic stage of anaerobic digestion (Vasconcelos et al., 2019). Furthermore, Clostridium lituseburense abundance was especially high in the acclimated inoculum treatments compared to the non-acclimated ones. C. lituseburense is known to hydrolyze benzene- and p-toluenesulfonates that can be further converted into methane in the anaerobic digestion trophic web (Shcherbakova et al., 2015). It also has been related to oxytetracycline resistance and its biodegradation under anaerobic conditions (Akyol et al., 2016). The compound p-toluenesulfonate is a cyclic organic compound with similar steric hindrances as cyclic siloxanes (Kino et al., 2021); hence, C. lituseburense ability to rapidly overcome steric hindrances may have contributed to D4 and D5 degradation, explaining the shorter lag times of the acclimated inoculum treatments.

In the microbial composition of the pre-BMP, there were notable differences in the Synergistales order between the non-acclimated and acclimated inocula (Fig. 2, and **supplementary material Fig. S1**). The non-acclimated inoculum had a high abundance of a HA73 Synergistales population, while the acclimated inoculum had a more balanced abundance of HA73 and *Aminobacterium* Synergistales populations (Fig. 2, [**A**]). Also, in the acclimated inoculum, the abundance of *D*ethiosulfovibronaceae spp., Rhodococcus spp., Tepidimicrobium spp., and Tissierella_Soehngenia spp. increased during the PDMS and microaeration acclimation process (Fig. 2, [A]). The beta diversity analyses showed ecological reshaping in all the anaerobic treatments during the BMP assay (**See supplementary material Fig. S2**). These analyses suggest that prior inoculum acclimation was the principal determinant of microbiome differentiation (as opposed to microaeration – or – D4:D5 supplementation during the BMP assay). The overall OTU composition differed significantly between microaeration and D4:D5 treatments (Fig. 2), but the abundance of the specific microbes, likely associated with D4:D5 degradation, differed only slightly (Fig. 1, Fig. 2).

Conversely, minor changes in the microbiome were observed in the non-acclimated inoculum treatments. Here, microbes such as *Dethiosulfuvibrio* spp. and *Peptostreptococcaceae* spp. decreased in abundance with an increase in D4:D5 concentration (Pearson correlation: p 0.178; r^2 0.465) (Fig. 2 [**B**, **C**, **D**]). On the other hand, the microbial composition of acclimated inoculum treatments showed no significant response to D4:D5 concentrations, suggesting microbiome adaptation to siloxanes compounds from prior PDMS supplementation. When microaeration was administered alongside D4:D5 supplementation, the microbiome composition between acclimated and non-acclimated treatments diverged substantially during the BMP assay (p 0.013; C.I.: 95 %). In general, microaeration had the effect of enlarging pre-existing differences in species abundance between acclimation treatments.

3.2. *Microaeration enhances D4:D5 conversion into methane, improves microbial kinetics, and modifies the microbiome*

Microaeration increased the conversion of D4:D5 into methane in all tested conditions, as seen when comparing anaerobic vs. microaerated treatments with D4:D5 supplementation (Fig. 1 [B0 vs. B1]). Methane production increased substantially in the 500 ppm D4:D5 treatments, exceeding the theoretical methane yield of the organic substrate (Fig. 1 [A1 to B2]) and their anaerobic counterparts by 7 to 11 % (p-value: 0.0254; C.I.: 95 %). The 500 ppm D4:D5 treatments also produced, on average, 57 % more methane than the control conditions (Fig. 1 [A0 & B0 vs. A1 & B2]). These results suggest that microaeration improved the hydrolysis of the organic substrate constituents and/or D4:D5, resulting in higher methane yields (Loughrin and Lovanh, 2019). Also, 3 % to 11

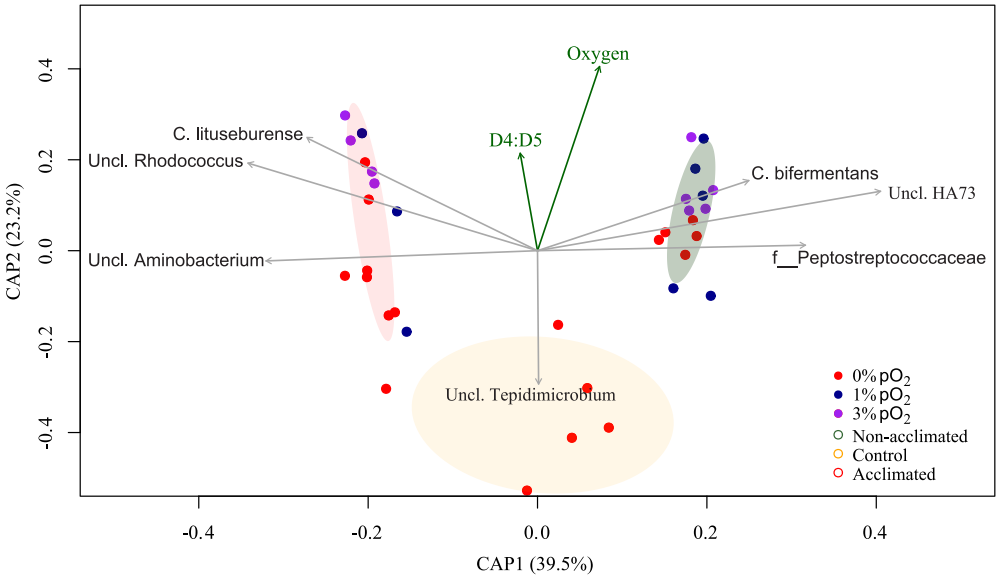


Fig. 3. Distance-based redundancy analysis from Bray-Curtis distance matrix. Green arrows show eigenvectors corresponding to the operative variables: oxygen partial pressure, D4:D5 concentration, and changes in β-diversity. Grey arrows indicate eigenvectors corresponding to the most significant microbes per condition and cluster. Clusters are represented by shaded ovals in the figure: non-acclimated inoculum (green), acclimated inoculum (red), and control conditions (yellow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

% (p-value: 0.0398; C.I.: 95 %) more methane was produced under microaerobic conditions compared to anaerobic conditions in the 50 ppm D4:D5 treatments (Fig. 1 [A0 & B0 vs. A1 & B2]). Moreover, the methane yield of the control treatments without D4:D5 supplementation increased slightly with microaeration (Fig. 1). However, there were no significant differences in methane yields between the non-acclimated and acclimated inocula for either the 1 % or 3 % P_{O_2} microaeration treatments (p-value: 0.538; C.I.: 95 %), indicating that microaeration, while advantageous, is not required for D4 or D5 biodegradation. Furthermore, high oxygen concentrations can slightly hinder methane production in non-acclimated treatments. For example, treatment T5_N (Fig. 1 **[A2]**) exhibited a lower methane yield compared to the control without D4:D5 supplementation. These results suggest that while microaeration may enhance the hydrolysis stages of anaerobic digestion (AD), there are still stochastic factors regarding how the microbiome reacts to high oxygen pressures ($P0_{O_2}$) that require further study and elucidation.

Previous studies link the improved methane yields during microaeration to an oxygen-driven reshaping of the AD microbiome that enhances the hydrolytic stage (Montalvo et al., 2018). Oxygen is thought to increase the abundance of facultative anaerobes (e.g., heterotrophs), thereby increasing the metabolic diversity of the reactor microbiome (Nguyen et al., 2019). These facultative microbes quickly utilize the supplied oxygen, maintaining low oxygen and redox potentials to protect oxygen-sensitive microbes (e.g., acetogens and methanogens) (Nguyen et al., 2019). Here, Tissierrella Soehngenia spp. and Tepidimicrobium spp. were more abundant in the pre-BMP acclimated inoculum, suggesting microaeration favored clostridial populations (Fig. 2, and **supplementary material Fig. S1**). Clostridia populations produce potent lipases, proteases, and esterases, that make them particularly effective in hydrolyzing lipids, proteins, and carboxylic acids (Damodara Kannan et al., 2020). For example, methane production rates increased by 38 % and 44 % in the treatments with acclimated inoculum, 50 ppm D4:D5 supplementation, and 1 % P_{O_2} and 3 % P_{O_2}, respectively (Table 1). The lag phase also decreased from 1 to 3 days compared to the anaerobic treatments. Previously, increased abundance of clostridia populations has been associated with siloxane biodegradation systems. Flores-Salgado *et al.* described a significant rise in *Clostridium* genus abundance within a two-stage system treating D4 and D5 siloxanes under anoxic conditions (Flores-Salgado et al., 2023). However, the volatile siloxanes biodegradation was attributed to *Mycobacterium* species as the primary responsible for D4 and D5 removal. Thus, the higher diversity and abundance of Clostridia may have contributed to the observed increase in kinetics, methane yields, and D4:D5 metabolism through enhanced hydrolytic processes, rather than directly transforming siloxanes molecules.

However, for the 500 ppm D4:D5 treatment, microaeration did not significantly improve kinetics, which may have been caused by inhibition from D4:D5 metabolites. Previous reports showed siloxane catabolism produced antimicrobial metabolites (i.e., phenol, aliphatic alcohols) that inhibited methane production at high initial siloxane concentrations (e.g., 500 ppm) (Ortiz-Ardila et al., 2021). These inhibitory effects may be concentration-dependent and, thus, only manifest in batch systems. If hydrolysis rates increase due to microaeration, the conversion rates of subsequent trophic steps, such as acetogenesis and methanogenesis, must increase proportionally to prevent an accumulation of intermediate products (Duarte et al., 2018). When microaeration was applied, the Synergistales order was more abundant in the acclimated treatments than the non-acclimated treatments (Fig. 2 [A vs. C vs. D]). When exposed to microaeration, HA73 Synergistales increased more than 60 % in abundance in the non-acclimated treatments but decreased by ~ 50 % in the acclimated treatments (Fig. 2 [A vs. C vs. D]). However, the inverse occurred for Aminobacterium spp. between treatments. In the non-acclimated inoculum, Aminobacterium spp. constituted less than 9 % of the community composition, whereas in the acclimated inoculum, they represented over 15 %.

Previous reports correlated Synergistales abundance under stress conditions (e.g., ammonia, high organic loading rate (OLR)) with improved hydrolysis rates and methane production (Militon et al., 2015). For example, Durán et al. (Durán et al., 2018) found that the abundance of Aminobacterium spp. significantly increased in an anaerobic membrane bioreactor (AnMBR) when the system was operated at a high organic loading rates (OLR) resulting in higher hydrolysis rates and methane production. Also, in an anaerobic system treating tuna cooking wastewater, with high ammonia content, HA73 Synergistales, which constituted 69 % relative abundance, were linked to peptone and amino acid degradation (Militon et al., 2015). HA73 Synergistales are syntrophic fermentative bacteria that have strong syntrophy with methanogenic archaea (Damodara Kannan et al., 2020), thus helping to maintain the balance between propionate, H_2, and acetate levels in anaerobic systems under stress (e.g., high OLR, high copper and sulphate concentrations) (Jordaan et al., 2019). These reports support the inferences on a microbial ecology re-balance when microaeration is applied for D4:D5 siloxanes conversion to methane.

The distance-based redundancy analysis (dbRDA) shows a strong correlation between methane production, oxygen addition, acclimated inoculum, and the microbial community composition (Fig. 3). The abundance of key species such as C. bifermentans, C. lituseburense, and Rhodococcus spp. increased in response to microaeration in the non-acclimated and acclimated treatments (Fig. 2, **and supplementary material Fig. S1**). These microbial populations have been related to the biodegradation of recalcitrant compounds under anaerobic conditions (Shcherbakova et al., 2015; Zhao et al., 2004), and some studies further suggest that microaeration can enhance their growth and metabolic capacities (Katarzyna et al., 2014). For example, the ability of C. bifermentans to degrade recalcitrant compounds has been shown to increase under microaerated conditions (Katarzyna et al., 2014; Myszka et al., 2012). Rather than intracellular adsorption, the degradation of complex molecules by C. bifermentans is mediated by extracellular enzymes with high redox capacity (Joe et al., 2008; Myszka et al., 2012).

The degradation rate of recalcitrant compounds by C. bifermentans depends on enzyme concentration and activity. Accordingly, a higher cell count (*i.e.,* abundance) yields higher amounts of these extracellular enzymes and, therefore, a higher activity. In this study, microaeration increased the abundance of C. bifermentans compared to anaerobic conditions in the non-acclimated treatments (Fig. 2 [B vs. C]). Myszka et al. (Myszka et al., 2012) suggested that *C. bifermentans* spores germinate faster in microaerated media than in anaerobic media, producing more metabolically active vegetative cells. Finally, the dbRDA analysis shows a strong correlation between P_{O_2}, D4:D5 concentrations, and C. bifermentans abundance (Fig. 3), suggesting that these microbes may have played a role in D4:D5 degradation.

C. lituseburense and Rhodococcus spp. also correlated with the D4:D5 conversion to methane (Fig. 3). The abundance of these microbes differed significantly between inocula treatments (Fig. 2 [B vs. C]). C. lituseburense abundance was > 5 % in all acclimated treatments and below detection limits in non-acclimated treatments (Fig. 2). In contrast, the abundance of Rhodococcus spp. was ~ 100 % higher in the acclimated treatments compared with non-acclimated ones (Fig. 2, **and supplementary material Fig. S1**). Rhodococcus spp. are known for their metabolic diversity, ecological adaptability (Kuyukina and Ivshina, 2019), and their ability to degrade recalcitrant compounds under microaerobic conditions (Fuller et al., 2010), including PDMS (Ortiz-Ardila et al., 2021). Supporting these results, members of Rhodococcus genus have also been identified in biotrickling filters utilized for treating volatile siloxanes. Specifically, *Rhodococcus* sp., *Rhodococcus erythropolis*, and *Rhodococcus qingshengii* were reported as part of a siloxane degradation community reported by Boada et al. (Boada et al., 2020) *R*hodococci are thought to participate in the oxidative catalysis of (poly) aromatic compounds using dioxygenase enzymes (Fahy et al., 2008). Furthermore, the abundance and metabolic capabilities of *Rhodococcus* spp. have been shown to increase under low-oxygen concentrations

(Fuller et al., 2010). This increase in abundance of *Rhodococcus* spp., which was also observed in the microaerated treatments of this study (Fig. 2 [B vs. C vs. D]), may have been correlated with the degradation D4:D5 and the associated increase in methane production (Fig. 3).

Finally, results show that microaerated conditions improved the conversion of D4:D5 into methane by restructuring the microbiome and improving the microbial kinetics. Additionally, the BMP assay results suggest that specific species use oxygen to overcome the steric hindrance of the D4 or D5 molecules. Previous reports suggest that the microbial degradation of volatile siloxanes might be an oxidative-enzymatic process that relies on oxygen as the final electron acceptor to catalyze the opening of cyclic siloxanes, allowing biodegradation (Li et al., 2014; Wang et al., 2014). Still, further studies should be conducted to elucidate the exact enzymatic mechanisms and microbial relationships that allow and enhance siloxane conversion into methane.

3.3. D4 biodegradation occurs as a stepwise process

Several siloxane-type compounds were detected in the microaeration treatments. Non-polar GC–MS runs detected: 1) octamethylcyclotetrasiloxane; 2) 1,1-dihydroxymethyl-2,2,3,3,4,4-octamethylcyclotetrasiloxane; and 3) 2,2,3,3,4,4-octamethyl cyclotetrasiloxan-1,1-dicarbaldehyde. Polar GC–MS runs additionally detected: 4) 2,2,3,3,4,4-octamethyl cyclotetrasiloxane-1,1-diol; and 5) 7-isobutoxy 1,1,3,3,5,5,7,7-octamethyl tetrasiloxan-1-ol. Finally, GC–MS runs with derivatized fractions detected: 6A) 1,1,3,3,5,5 hexamethyltrisiloxane; 6B) octamethyltrisiloxane; and 7) hexamethyldisiloxane (Fig. 4).

The theoretical reconstruction of the D4 biodegradation route under microaerated conditions proposes a stepwise pathway that first overcomes the steric hindrance, which protects the siloxane ring from opening. This siloxane steric hindrance has been known to confer stability to the D4 molecule and be responsible for the physicochemical characteristics and recalcitrant nature of volatile siloxanes (Ortiz-Ardila et al., 2023). Previous studies have neglected the siloxane steric hindrance, proposing a D4 biodegradation pathway that starts at the siloxane-ring opening by the microbial catalysis of the Si-O bond (Li et al., 2014). In the proposed pathway, the difficulty of cleaving the Si-O bond is recognized as a 4-step enzyme-mediated process (Fig. 4 Molecules **1** to 5). The presence of molecules **2** and **3** suggests that the D4 biodegradation began with removing the shielding effect of the methyl groups attached to the silicon molecule by oxidating them to primary alcohols.

The oxidation of the methyl groups responsible for the steric hindrance was identified by the change in the molecular mass of the D4 molecule from 296 Da to 328 Da (Fig. 4 molecules **1** & **2**), corresponding to the replacement of two H^+ with OH^- groups. The GC–MS analysis

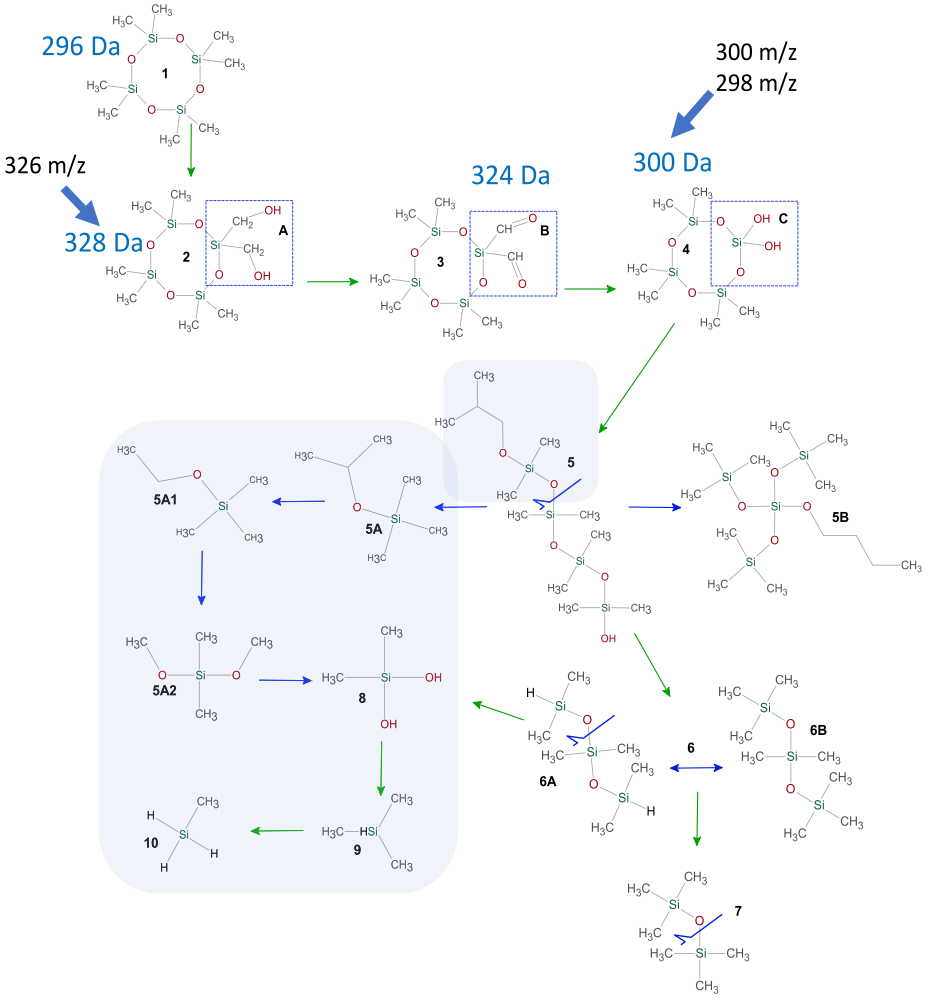


Fig. 4. Possible pathways of D4 biodegradation under microaerated conditions in the BMP assays. Numbers in blue refer to the molecule weight in Daltons (Da). Boxes with dashed outlines highlight the functional groups participating in the oxidation reaction and frame the mass spectra peak fragment that confirms the structure. Numbers in black with $^{m}/_{z}$ symbols correspond to the mass peak identifying each molecule. Green arrows show the general biodegradation pathway. Blue arrows indicate alternative pathways from the marked precursors (Shaded blue background). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

identified a siloxane molecule with 326 $^{m}/_{z}$ as the mass peak, and the fragmentation pattern suggests the presence of methyl-oxidized groups. The D4 molecule mass-fragmentation pattern confirmed the attachment of these methyl-oxidized groups to the SI-O bond. Here, the 107 $^{m}/_{z}$ mass peak was related to the Si-CH_2-OH fraction in molecule 2, indicating that the D4 siloxane methyl substituents were oxidized (Fig. 4 molecule 2 – A). The results suggest that facultative anaerobes might remove the steric hindrance on siloxane molecules to allow their stepwise biodegradation. Wang et al. (Wang et al., 2014) reported the same initial biodegradation process but under aerobic conditions, asserting that the oxidation of methyl groups in the siloxane molecule is only possible when a strong final electron acceptor (*i.e.,* oxygen) is available.

Once terminal alcohols replace the methyl groups (Fig. 4 molecule 2), microbes can oxidize them to aldehydes (Fig. 4 molecule 3) and derive energy. Molecule **3** was identified by the mass peak at 324 Da and the mass fractions 57 $^{m}/_{z}$, 73 $^{m}/_{z}$, and 85 $^{m}/_{z}$, showing the oxidation from the Si-CH_2-OH to the aldehyde substituent Si-CH-O. These stepwise reactions involving microbial metabolism have never been described for siloxane molecules. The reaction products suggest the presence of dehydrogenase enzymes that use the NAD^+ aerobic system to catalyze alcohol oxidation. Subsequent reaction steps suggest a possible nucleophilic attack on the aldehydes carbonyl center that removes the carbonyl group and substitutes the –OH radical (Fig. 4 molecule **4**). Molecule **4** was identified by mass peaks 300 $^{m}/_{z}$, 298 $^{m}/_{z}$, and mass fraction 94 $^{m}/_{z}$, which confirm the Si-OH bond. The resultant alcohol attached to the Si atom is subject to electron repulsion between its terminal –OH free electrons and the oxygen free-electron pairs in the Si-O-Si cyclic structure. This repulsion and the availability of free electrons provide a target area for nucleophilic reactions on the Si-O-Si structure that opens the cyclic configuration.

Previous D4 and D5 degradation studies report that once the siloxane ring is open, catalysis of the linear molecule proceeds stepwise (Accettola et al., 2008; Zhang et al., 2020), breaking the molecule down one monomer at a time (Li et al., 2014). The mass spectra results identified the cleavage product of the octamethyltetrasiloxanol molecule, supporting this explanation (Fig. 4 molecule **5**). Once the D4 molecule is open, the OH terminals become susceptible to further reactions with other compounds. For example, the isobutyl substituent in molecule 5 (Fig. 4 molecule **5**) might result from forming an ether bond between the linear siloxane molecule and the metabolite, isobutanol. Other studies support these inferences, showing that alkyl siloxanes or siloxane molecules with long organic substituents, such as molecule **5B**, are more susceptible to biodegradation once other organic substituents are attached (Rościszewski et al., 1998).

On the other hand, trimethylsiloxane **5A**, as a by-product, might be released after the Si-O bond is broken in molecule **5** and further catalyzed by the demethylation of the isobutyl substituent (Fig. 4 molecule **5A1**). This reaction has been previously described in *Rhodococcus* sp. (Venkatesagowda and Dekker, 2021). *Rhodococcus* spp. performs isobutyl demethylation using oxygen-dependent demethylases that transform alkyl products into simpler molecules that can be further metabolized (Venkatesagowda and Dekker, 2021). Other species have also been reported to perform demethylation reactions in anaerobic digestion microbiomes. For example, Boada *et al.* reported that *Methylibium* spp. use methylotrophic reactions to obtain energy from siloxanes for metabolism and growth (Boada et al., 2020). The elucidation of chemical reactions suggests D4 was similarly degraded to CO_2 biologically through the oxidation of alkyl substituents (*i.e.,* butyl, methyl, ethyl), which yielded energy for growth or metabolism. The identification of methyl ether (molecule **5A2**), which might be produced by the demethylation and oxidation of the by-product ethoxy(trimethyl)silane (**5A1**), supports this explanation. As previously reported, dimethoxy (dimethyl)silane (**5A2**) can be converted into dimethylsilanediol (DMSD) (Fig. 4 molecule **8**) as one of the final products of D4 or D5 biodegradation (Ortiz-Ardila et al., 2023; Zhang et al., 2020).

In addition, L3 siloxane molecules (Fig. 4 molecules **6A**, **6B**) that were not present at the beginning of the BMP assay, nor the controls without siloxanes, were identified in the post-BMP samples. The presence of L3 molecules indicates D4 or D5 was degraded. It is inferred that these L3 molecules were metabolized to DMSD by cleaving one siloxane monomer at a time, as suggested by Li *et al.* (Li et al., 2014). This metabolic process is mediated by oxidation reactions that convert the methyl substituents into Si-O-CH_3 ethers, which are further transformed into methanol and CO_2 (Li et al., 2014). The methanol and CO_2 may have been used by methylotrophic and hydrogenotrophic methanogens, explaining the increased methane production in the BMP assays. The proposed pathway shares multiple similarities with the previously reported D4 aerobic biodegradation pathway (Wang et al., 2014), suggesting microaerobic conditions were bioenergetically sufficient to enable this pathway. The body of results presented here strongly suggests volatile siloxanes can be biodegraded, and their by-products can be converted to methane in a methanogenic-microaerobic system, demonstrating that aerobic/anaerobic biochemical reactions can co-occur in a single reactor-developed microbiome.

3.4. Microaeration as a practical strategy to reduce volatile siloxanes in anaerobic digestion

Microaeration has been tested extensively at the lab scale since the 2000 s to enhance organic matter biodegradability, methane production, and hydrogen sulfide (H_2S) removal (Chen et al., 2020). Initial reports showed a significant enhancement of over 40–60 % in methane production under microaerated conditions compared with anaerobic controls (Chen et al., 2020). Microaeration has been upscaled to ~ 200 L pilot- and ~ 400 m^3 full-scale capacities using various bioreactor configurations and substrates (e.g., primary sludge, wastewater, sewage sludge) (Nguyen et al., 2019). Microaeration is a well-known and robust strategy for removing H_2S from biogas during anaerobic digestion while improving COD removal (Nguyen and Khanal, 2018). Furthermore, multiple studies have explored using microaeration to shape anaerobic digestion microbiomes for improved performance (Chen et al., 2020; Menezes et al., 2019).

Despite the considerable research and developmental effort around microaeration, full-scale applications outside biogas desulfurization are limited. The application of microaeration to biodegrade recalcitrant compounds is still in its early stages (Chen et al., 2020). This work and other laboratory-scale experiences show that microaeration can improve the biodegradation of recalcitrant or toxic compounds, such as siloxanes, phenols, furans, sulfonates, and azo dyes (Chen et al., 2020; Menezes et al., 2019). However, mechanistic explanations of how microaeration promotes the biodegradation of complex substrates, shapes the microbiome, and improves microbial kinetics are lacking and require further study. This work contributes to these explanations by demonstrating how microaerated conditions shape the microbiome to enable methane production from volatile siloxanes and by providing insights into the dynamics of D4 biodegradation. Furthermore, this study intends to provide an initial proof of concept to encourage other researchers to test and scale microaeration strategies for treating volatile siloxanes and other recalcitrant pollutants.

4. Conclusions

- D4:D5 were biodegraded and converted into methane under anaerobic conditions using acclimated and non-acclimated inocula.
- Oxygen partial pressures of ~ 3 %P_{O_2} improved biodegradation of D4:D5 and increased methane yields.
- Microaeration reshaped the anaerobic microbiome, promoting facultative species enrichment and improving process kinetics.
- D4 biodegradation under microaerated conditions likely involved a stepwise process that overcame the steric hindrances by first oxidizing the siloxane ring methyl groups.

CRediT authorship contribution statement

A.E. Ortiz-Ardila: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Visualization, Writing – original draft. **C. Celis:** Formal analysis, Investigation. **J.G. Usack:** Funding acquisition, Writing – review & editing. **L.T. Angenent:** Methodology, Resources, Supervision, Validation, Writing – review & editing. **R.A. Labatut:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was partially funded by the Chilean research and development agency (ANID) under the 2017 National PhD Scholarship #21170252 and the FONDECYT Project #1242040. Also, by the Federal Ministry of Education and Research (BMBF) and the Baden-Württemberg Ministry of Science as part of the Excellence Strategy of the German Federal and State Governments under project: PRO-ORTIZ-2022-15.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.130673.

References

Accettola, F., Guebitz, G.M., Schoeftner, R., 2008. Siloxane removal from biogas by biofiltration: Biodegradation studies. Clean Techn. Environ. Policy 10, 211–218. https://doi.org/10.1007/s10098-007-0141-4.

Akyol, Ç., Aydin, S., Ince, O., Ince, B., 2016. A comprehensive microbial insight into single-stage and two-stage anaerobic digestion of oxytetracycline-medicated cattle manure. Chem. Eng. J. 303, 675–684. https://doi.org/10.1016/j.cej.2016.06.006.

American Public Health Association, American Water Works Association, Water Environment Federation, 2017. Standard Methods for the Examination of Water and Wastewater, 23rd ed. APHA Press, Washington DC. doi: 10.2105/SMWW.2882.216.

Andersen, K.S., Kirkegaard, R.H., Karst, S.M., Albertsen, M., 2018. ampvis2: an R package to analyse and visualise 16S rRNA amplicon data. BioRxiv 299537.

Boada, E., Santos-Clotas, E., Bertran, S., Cabrera-Codony, A., Martín, M.J., Bañeras, L., Gich, F., 2020. Potential use of Methylibium sp. as a biodegradation tool in organosilicon and volatile compounds removal for biogas upgrading. Chemosphere 240. https://doi.org/10.1016/j.chemosphere.2019.124908.

Bolyen, E., Rideout, J.R., Dillon, M.R., Bokulich, N.A., et al., J.G., 2019. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nature Biotechnology 2019 37:8 37, 852–857. doi: 10.1038/s41587-019-0209-9.

Bonakdari, H., Zeynoddin, M., 2022. In: Stochastic Modeling. Elsevier. https://doi.org/10.1016/C2021-0-01534-0.

Chang, Y.C., Okeke, B.C., Hatsu, M., Takamizawa, K., 2001. In vitro dehalogenation of tetrachloroethylene (PCE) by cell-free extracts of Clostridium bifermentans DPH-1. Bioresour. Technol. 78, 141–147. https://doi.org/10.1016/S0960-8524(01)00005-0.

Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2020. Review on microaeration-based anaerobic digestion: State of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388 https://doi.org/10.1016/J.SCITOTENV.2019.136388.

Damodara Kannan, A., Evans, P., Parameswaran, P., 2020. Long-term microbial community dynamics in a pilot-scale gas sparged anaerobic membrane bioreactor treating municipal wastewater under seasonal variations. Bioresour. Technol. 310, 123425 https://doi.org/10.1016/j.biortech.2020.123425.

Duarte, M.S., Silva, S.A., Salvador, A.F., Cavaleiro, A.J., Stams, A.J.M., Alves, M.M., Pereira, M.A., 2018. Insight into the Role of Facultative Bacteria Stimulated by Microaeration in Continuous Bioreactors Converting LCFA to Methane. Environ. Sci. Tech. 52, 6497–6507. https://doi.org/10.1021/acs.est.8b00894.

Durán, F., Zamorano-López, N., Barat, R., Ferrer, J., Aguado, D., 2018. Understanding the performance of an AnMBR treating urban wastewater and food waste via model simulation and characterization of the microbial population dynamics. Process Biochem. 67, 139–146. https://doi.org/10.1016/j.procbio.2018.02.010.

European Chemical Agency, G., 2018b. Proposal for Identification of a Substance of Very High Concern on the Basis of the Criteria Set out in Article 57 (Octamethylcyclotetrasiloxane, D4).

European Chemical Agency, G., 2018a. Annex XV report Proposal for Identification of a Substance of Very High Concern on the Basis of the Criteria Set out in Article 57 (Decamethylcyclopentasiloxane; D5). Germany.

Fahy, A., Ball, A.S., Lethbridge, G., McGenity, T.J., Timmis, K.N., 2008. High benzene concentrations can favour Gram-positive bacteria in groundwaters from a contaminated aquifer. FEMS Microbiol. Ecol. 65, 526–533. https://doi.org/10.1111/J.1574-6941.2008.00518.X.

Flores-Salgado, G., Contreras, J.A., Pérez-Trevilla, J., Quijano, G., 2023. Removal of volatile methyl siloxanes in an anoxic two-phase partitioning bioreactor operated with hydrophobic biomass. Chem. Eng. J. 477 https://doi.org/10.1016/j.cej.2023.146985.

Fuller, M.E., Perreault, N., Hawari, J., 2010. Microaerophilic degradation of hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) by three Rhodococcus strains. Lett. Appl. Microbiol. 51, 313–318. https://doi.org/10.1111/j.1472-765X.2010.02897.x.

Gaj, K., 2020. Adsorptive biogas purification from siloxanes-A critical review. Energies (basel) 13. https://doi.org/10.3390/en13102605.

Hathaway, J.J.M., Moser, D.P., Blank, J.G., Northup, D.E., 2021. A Comparison of Primers in 16S rRNA Gene Surveys of Bacteria and Archaea from Volcanic Caves. Geomicrobiol J. 38, 741–754. https://doi.org/10.1080/01490451.2021.1943727.

Joe, M.-H., Lim, S.-Y., Kim, D.-H., Lee, I.-S., 2008. Decolorization of reactive dyes by Clostridium bifermentans SL186 isolated from contaminated soil. World J. Microbiol. Biotechnol. 24, 2221–2226. https://doi.org/10.1007/s11274-008-9733-3.

Jordaan, E.M., Levin, D.B., Sparling, R., Khafipour, E., Çiçek, N., 2019. Microbial population change in anaerobic digestion during copper sulfate inhibition and recovery. Am. Soc. Agric. Biol. Eng. 62, 1231–1241. https://doi.org/10.13031/trans.13462.

Katarzyna, L., Kamila, M., Agnieszka, K.-O.-S., Wojciech, J., Katarzyna, C., 2014. Selection and characterization of Clostridium bifermentans strains from natural environment capable of producing 1,3-propanediol under microaerophilic conditions. Afr. J. Microbiol. Res. 8, 1187–1197. https://doi.org/10.5897/ajmr213.6516.

Kino, D., Okada, K., Tokudome, Y., Takahashi, M., Malfatti, L., Innocenzi, P., 2021. Reactivity of silanol group on siloxane oligomers for designing molecular structure and surface wettability. J Solgel Sci. Technol. 97, 734–742. https://doi.org/10.1007/s10971-020-05448-z.

Kuyukina, M.S., Ivshina, I.B., 2019. Application of Rhodococcus in Bioremediation of Contaminated Environments. pp. 231–262. doi: 10.1007/978-3-642-12937-7_9.

Li, Y., Zhang, W., Xu, J., 2014. Siloxanes removal from biogas by a lab-scale biotrickling filter inoculated with Pseudomonas aeruginosa S240. J. Hazard. Mater. 275, 175–184. https://doi.org/10.1016/j.jhazmat.2014.05.008.

Loughrin, J., Lovanh, N., 2019. Aeration to improve biogas production by recalcitrant feedstock. Environments - MDPI 6, 4–13. https://doi.org/10.3390/environments6040044.

Menezes, O., Brito, R., Hallwass, F., Florêncio, L., Kato, M.T., Gavazza, S., 2019. Coupling intermittent micro-aeration to anaerobic digestion improves tetra-azo dye Direct Black 22 treatment in sequencing batch reactors. Chem. Eng. Res. Des. 146, 369–378. https://doi.org/10.1016/j.cherd.2019.04.020.

Militon, C., Hamdi, O., Michotey, V., Fardeau, M.L., Ollivier, B., Bouallagui, H., Hamdi, M., Bonin, P., 2015. Ecological significance of Synergistetes in the biological treatment of tuna cooking wastewater by an anaerobic sequencing batch reactor. Environ. Sci. Pollut. Res. 22, 18230–18238. https://doi.org/10.1007/s11356-015-4973-x.

Mojsiewicz-Pieńkowska, K., Krenczkowska, D., 2018. Evolution of consciousness of exposure to siloxanes—review of publications. Chemosphere 191, 204–217. https://doi.org/10.1016/j.chemosphere.2017.10.045.

Montalvo, S., Vielma, S., Borja, R., Huiliñir, C., Guerrero, L., 2018. Increase in biogas production in anaerobic sludge digestion by combining aerobic hydrolysis and addition of metallic wastes. Renew. Energy 123, 541–548. https://doi.org/10.1016/j.renene.2018.02.004.

Myszka, K., Leja, K., Olejnik-Schmidt, A.K., Czaczyk, K., 2012. Isolation process of industrially useful Clostridium bifermentans from natural samples. J. Biosci. Bioeng. 113, 631–633. https://doi.org/10.1016/j.jbiosc.2012.01.003.

Nguyen, D., Khanal, S.K., 2018. A little breath of fresh air into an anaerobic system: How microaeration facilitates anaerobic digestion process. Biotechnol. Adv. 36, 1971–1983. https://doi.org/10.1016/j.biotechadv.2018.08.007.

Nguyen, D., Wu, Z., Shrestha, S., Lee, P.H., Raskin, L., Khanal, S.K., 2019. Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res. 166, 115080 https://doi.org/10.1016/j.watres.2019.115080.

Ortiz-Ardila, A.E., Díez, B., Celis, C., Jenicek, P., Labatut, R., 2021. Microaerobic conditions in anaerobic sludge promote changes in bacterial composition favouring biodegradation of polymeric siloxanes. Environ. Sci. Process Impacts 23, 1182–1197. https://doi.org/10.1039/d1em00143d.

Ortiz-Ardila, A.E., Restrepo, J.D., Angenent, L.T., Usack, J.G., Labatut, R.A., 2023. Protecting human health and the environment against siloxanes: The role and effectiveness of wastewater treatment technologies. Crit. Rev. Environ. Sci. Technol. 1–27 https://doi.org/10.1080/10643389.2023.2221156.

Piechota, G., Buczkowski, R., 2014. Development of chromatographic methods by using direct-sampling procedure for the quantification of cyclic and linear volatile methylsiloxanes in biogas as perspective for application in online systems. Int. J. Environ. Anal. Chem. 94, 837–851. https://doi.org/10.1080/03067319.2013.879296.

R Core Team, 2021. R: A language and environment for statistical computing (Version 4.0. 5)[Computer software]. R Foundation for Statistical Computing.

Rościszewski, P., Łukasiak, J., Dorosz, A., Galiński, J., Szponar, M., 1998. Biodegradation of Polyorganosiloxanes. Macromol Symp 130, 337–346. https://doi.org/10.1002/MASY.19981300129.

Sanchís, J., Martínez, E., Ginebreda, A., Farré, M., Barceló, D., 2013. Occurrence of linear and cyclic volatile methylsiloxanes in wastewater, surface water and sediments from Catalonia. Sci. Total Environ. 443, 530–538. https://doi.org/10.1016/j.scitotenv.2012.10.047.

Santos-Clotas, E., Cabrera-Codony, A., Boada, E., Gich, F., Muñoz, R., Martín, M.J., 2019. Efficient removal of siloxanes and volatile organic compounds from sewage biogas by an anoxic biotrickling filter supplemented with activated carbon. Bioresour. Technol. 294, 122136 https://doi.org/10.1016/j.biortech.2019.122136.

Santos-Clotas, E., Cabrera-Codony, A., Comas, J., Martín, M.J., 2020. Biogas purification through membrane bioreactors: Experimental study on siloxane separation and biodegradation. Sep. Purif. Technol. 238, 116440 https://doi.org/10.1016/j.seppur.2019.116440.

Shcherbakova, V.A., Laurinavichyus, K.S., Chuvil'skaya, N.A., Ryzhmanova, Y. V, Akimenko, V.K., 2015. Anaerobic bacteria involved in the degradation of aromatic sulfonates to methane. Appl Biochem Microbiol 51, 209–214. doi: 10.1134/S0003683815020179.

Struk, M., Kushkevych, I., Vítězová, M., 2020. Biogas upgrading methods: recent advancements and emerging technologies. Rev. Environ. Sci. Biotechnol. 19, 651–671. https://doi.org/10.1007/s11157-020-09539-9.

Symons, G.E., Buswell, A.M., 1933. The Methane Fermentation of Carbohydrates. J. Am. Chem. Soc. 55, 2028–2036. https://doi.org/10.1021/ja01332a039.

Tjørve, K.M.C., Tjørve, E., 2017. The use of Gompertz models in growth analyses, and new Gompertz-model approach: An addition to the Unified-Richards family. PLoS One 12, 1–17. https://doi.org/10.1371/journal.pone.0178691.

van Lier, J.B., Mahmoud, N., Zeeman, G., 2023. Anaerobic wastewater treatment. Biological Wastewater Treatment: Principles, Modeling and Design 701–756. https://doi.org/10.2166/9781789060362_0701.

Vasconcelos, E.A.F., Santaella, S.T., Viana, M.B., dos Santos, A.B., Pinheiro, G.C., Leitão, R.C., 2019. Composition and ecology of bacterial and archaeal communities in anaerobic reactor fed with residual glycerol. Anaerobe 59, 145–153. https://doi.org/10.1016/j.anaerobe.2019.06.014.

Venkatesagowda, B., Dekker, R.F.H., 2021. Microbial demethylation of lignin: Evidence of enzymes participating in the removal of methyl/methoxyl groups. Enzyme Microb. Technol. 147, 109780 https://doi.org/10.1016/j.enzmictec.2021.109780.

Wang, J., Zhang, W., Xu, J., Li, Y., Xu, X., 2014. Octamethylcyclotetrasiloxane removal using an isolated bacterial strain in the biotrickling filter. Biochem. Eng. J. 91, 46–52. https://doi.org/10.1016/j.bej.2014.07.003.

Wang, G., Zhang, Z., Hao, Z., 2019. Recent advances in technologies for the removal of volatile methylsiloxanes: A case in biogas purification process. Crit. Rev. Environ. Sci. Technol. 49, 2257–2313. https://doi.org/10.1080/10643389.2019.1607443.

Wu, Z., Nguyen, D., Lam, T.Y.C., Zhuang, H., Shrestha, S., Raskin, L., Khanal, S.K., Lee, P. H., 2021. Synergistic association between cytochrome bd-encoded Proteiniphilum and reactive oxygen species (ROS)-scavenging methanogens in microaerobic-anaerobic digestion of lignocellulosic biomass. Water Res. 190, 116721 https://doi.org/10.1016/j.watres.2020.116721.

Xu, Q., Vandenkoornhuyse, P., Li, L., Guo, J., Zhu, C., Guo, S., Ling, N., Shen, Q., 2022. Microbial generalists and specialists differently contribute to the community diversity in farmland soils. J. Adv. Res. 40, 17–27. https://doi.org/10.1016/j.jare.2021.12.003.

Zhang, Y., Oshita, K., Kusakabe, T., Takaoka, M., Kawasaki, Y., Minami, D., Tanaka, T., 2020. Simultaneous removal of siloxanes and H2S from biogas using an aerobic biotrickling filter. J. Hazard. Mater. 391, 122187 https://doi.org/10.1016/j.jhazmat.2020.122187.

Zhao, J.-S., Paquet, L., Halasz, A., Manno, D., Hawari, J., 2004. Metabolism of octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazocine by Clostridium bifermentans strain HAW-1 and several other H2-producing fermentative anaerobic bacteria. FEMS Microbiol. Lett. 237, 65–72. https://doi.org/10.1111/J.1574-6968.2004.TB09679.X.